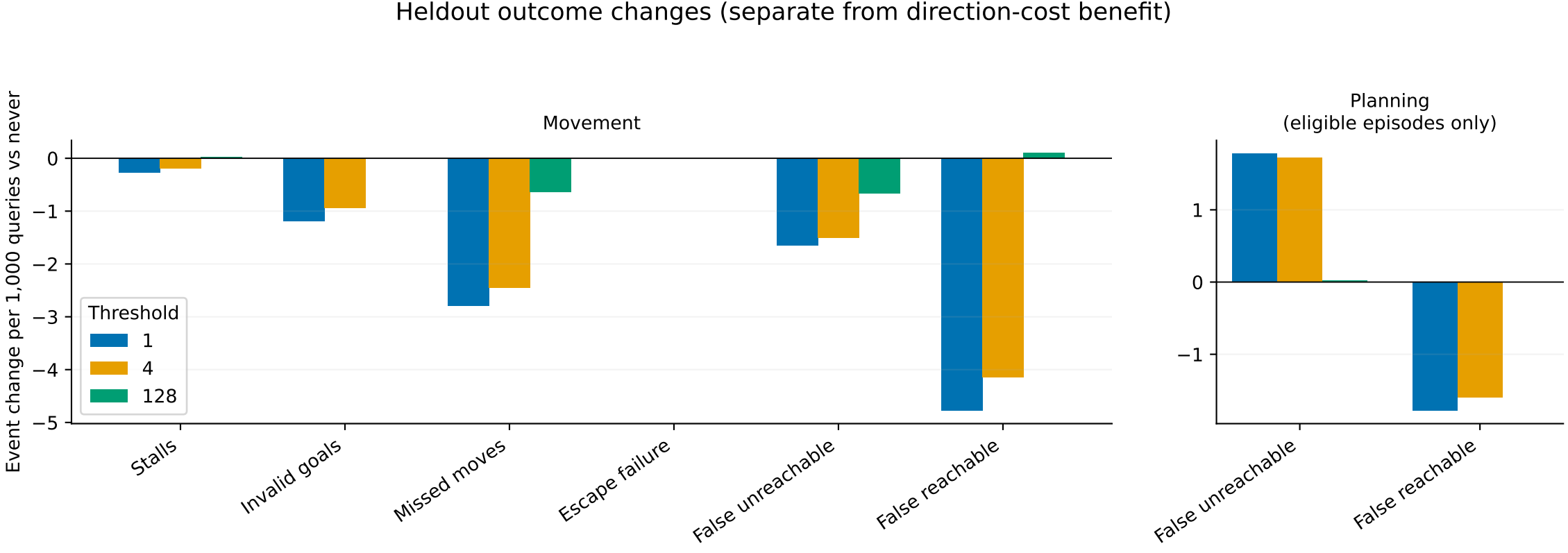
Heldout outcome changes (separate from direction-cost benefit)
Movement
Planning
(eligible episodes only)
Event change per 1,000 queries vs never
0
−1
−2
−3
−4
−5
1
0
−1
Threshold
1
4
128
Stalls
Invalid goals
Missed moves
Escape failure
False unreachable
False reachable
False unreachable
False reachable
Negative = fewer events. Offline overlapping counterfactual episodes, not whole-game outcomes; no CPU conversion.